

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\polished_dataset ;
- comparison mode: full_directional_candidate_matrix ;
- candidate count: 166 ;
- held-out curve count before candidate filtering: 194 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;
- global candidates are evaluated on both directions and reported with direction-separated metrics.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Directions
SVM19_Fw	SVM	rcim_track1	track1_reference_bank	Fw	forward
SVM19_Bw	SVM	rcim_track1	track1_reference_bank	Bw	backward
MLP19_Fw	MLP	rcim_track1	track1_reference_bank	Fw	forward
MLP19_Bw	MLP	rcim_track1	track1_reference_bank	Bw	backward
RF19_Fw	RF	rcim_track1	track1_reference_bank	Fw	forward
RF19_Bw	RF	rcim_track1	track1_reference_bank	Bw	backward
DT19_Fw	DT	rcim_track1	track1_reference_bank	Fw	forward
DT19_Bw	DT	rcim_track1	track1_reference_bank	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
ET19_Fw	ET	rcim_track1	track1_reference_bank	Fw	forward
ET19_Bw	ET	rcim_track1	track1_reference_bank	Bw	backward
ERT19_Fw	ERT	rcim_track1	track1_reference_bank	Fw	forward
ERT19_Bw	ERT	rcim_track1	track1_reference_bank	Bw	backward
GBM19_Fw	GBM	rcim_track1	track1_reference_bank	Fw	forward
GBM19_Bw	GBM	rcim_track1	track1_reference_bank	Bw	backward
HGBM19_Fw	HGBM	rcim_track1	track1_reference_bank	Fw	forward
HGBM19_Bw	HGBM	rcim_track1	track1_reference_bank	Bw	backward
XGBM19_Fw	XGBM	rcim_track1	track1_reference_bank	Fw	forward
XGBM19_Bw	XGBM	rcim_track1	track1_reference_bank	Bw	backward
LGBM19_Fw	LGBM	rcim_track1	track1_reference_bank	Fw	forward
LGBM19_Bw	LGBM	rcim_track1	track1_reference_bank	Bw	backward
ELM19_Fw	ELM	rcim_track1	track1_reference_bank	Fw	forward
ELM19_Bw	ELM	rcim_track1	track1_reference_bank	Bw	backward
rcim_original_SVM19_Fw	SVM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_MLP19_Fw	MLP	rcim_original	track1_reference_bank	Fw	forward
rcim_original_RF19_Fw	RF	rcim_original	track1_reference_bank	Fw	forward
rcim_original_DT19_Fw	DT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ET19_Fw	ET	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ERT19_Fw	ERT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_GBM19_Fw	GBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_HGBM19_Fw	HGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_XGBM19_Fw	XGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_LGBM19_Fw	LGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ELM19_Fw	ELM	rcim_original	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Fw	SVM	rcim_retuned	track1_reference_bank	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
rcim_retuned_MLP19_Fw	MLP	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_RF19_Fw	RF	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_DT19_Fw	DT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ET19_Fw	ET	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ERT19_Fw	ERT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_GBM19_Fw	GBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_HGBM19_Fw	HGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_XGBM19_Fw	XGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_LGBM19_Fw	LGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ELM19_Fw	ELM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Bw	SVM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_MLP19_Bw	MLP	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_RF19_Bw	RF	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_DT19_Bw	DT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ET19_Bw	ET	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ERT19_Bw	ERT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_GBM19_Bw	GBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_HGBM19_Bw	HGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_XGBM19_Bw	XGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_LGBM19_Bw	LGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ELM19_Bw	ELM	rcim_retuned	track1_reference_bank	Bw	backward
temporal_convolution_global	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
temporal_convolution_Fw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
temporal_convolution_Bw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
gru_sequence_global	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
gru_sequence_Fw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
gru_sequence_Bw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
lstm_sequence_global	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
lstm_sequence_Fw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
lstm_sequence_Bw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
feedforward_global	feedforward	wave1	wave1_exported_model	global	forward, backward
feedforward_Fw	feedforward	wave1	wave1_exported_model	Fw	forward
feedforward_Bw	feedforward	wave1	wave1_exported_model	Bw	backward
harmonic_regression_global	harmonic_regression	wave1	wave1_exported_model	global	forward, backward
harmonic_regression_Fw	harmonic_regression	wave1	wave1_exported_model	Fw	forward
harmonic_regression_Bw	harmonic_regression	wave1	wave1_exported_model	Bw	backward
periodic_mlp_global	periodic_mlp	wave1	wave1_exported_model	global	forward, backward
periodic_mlp_Fw	periodic_mlp	wave1	wave1_exported_model	Fw	forward
periodic_mlp_Bw	periodic_mlp	wave1	wave1_exported_model	Bw	backward
residual_harmonic_mlp_global	residual_harmonic_mlp	wave1	wave1_exported_model	global	forward, backward
residual_harmonic_mlp_Fw	residual_harmonic_mlp	wave1	wave1_exported_model	Fw	forward
residual_harmonic_mlp_Bw	residual_harmonic_mlp	wave1	wave1_exported_model	Bw	backward
tree_global	tree	wave1	wave1_exported_model	global	forward, backward
tree_Fw	tree	wave1	wave1_exported_model	Fw	forward
tree_Bw	tree	wave1	wave1_exported_model	Bw	backward
paper_original_best_Fw	best_composite	rcim_original	composite_reference_bank	Fw	forward
paper_retuned_best_Fw	best_composite	rcim_retuned	composite_reference_bank	Fw	forward
track1_best_Fw	best_composite	rcim_track1	composite_reference_bank	Fw	forward
paper_retuned_best_Bw	best_composite	rcim_retuned	composite_reference_bank	Bw	backward
track1_best_Bw	best_composite	rcim_track1	composite_reference_bank	Bw	backward
periodic_temporal_convolution_global	periodic_temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
periodic_temporal_convolution_Fw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_temporal_convolution_Bw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_gru_sequence_global	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_gru_sequence_Fw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_gru_sequence_Bw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_lstm_sequence_global	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_lstm_sequence_Fw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_lstm_sequence_Bw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
sequential_residual_offset_probe_global	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_clean_sequential_residual_offset_global	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_harmonic_residual_offset_global	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_pointwise_control_global	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_centered_shape_global	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_offset_global	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_full_curve_composite_global	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2h_mae_robust_global	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_smooth_l1_robust_global	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
track2h_log_cosh_robust_global	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_quantile_p10_p50_p90_global	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_gaussian_nll_global	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k2_global	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k3_global	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_l_gru_offset_residual_global	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_causal_tcn_offset_residual_global	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_pointwise_control_global	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_smooth_l1_structured_global	wave3_harmonic_prior_residual_smooth_l1_structured	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave52b_offset_centered_shape_harmonic_global	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	global	forward, backward
polished_feedforward_global	feedforward	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_harmonic_regression_global	harmonic_regression	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_global	periodic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_mlp_global	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_tree_global	tree	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_harmonic_global	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_temporal_convolution_global	temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_gru_sequence_global	gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_lstm_sequence_global	lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_temporal_convolution_global	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_gru_sequence_global	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_lstm_sequence_global	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_1_sequential_residual_offset_probe_global	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_clean_sequential_residual_offset_global	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_harmonic_residual_offset_global	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_curve_aware_pointwise_control_global	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_centered_shape_curve_aware_global	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_offset_curve_aware_global	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave3_3_full_curve_composite_global	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_mae_robust_loss_global	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_smooth_l1_robust_loss_global	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_log_cosh_robust_loss_global	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_quantile_p10_p50_p90_global	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_gaussian_nll_global	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k2_global	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k3_global	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_gru_latent_offset_residual_global	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_causal_tcn_latent_offset_residual_global	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_pointwise_control_global	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	global	forward, backward

Best Composite Reference Models

These candidates combine the approved best harmonic-wise cells into one TE Curve Verification Pipeline curve-reconstruction candidate. They are also repeated inside the source-group tables below, but this section keeps the composed models explicit.

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
paper_original_best_Fw	rcim_original	Fw	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	Fw	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	Fw	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	Bw	backward	0.003675	0.004284	7.572	15.645

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track1_best_Bw	rcim_track1	Bw	backward	0.005027	0.005212	11.860	48.106

Forward Comparison

Original Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_original_ERT19_Fw	0.001471	0.001677	3.253	6.145
rcim_original_RF19_Fw	0.001767	0.001971	3.940	6.872
rcim_original_LGBM19_Fw	0.001801	0.002004	4.017	10.054
rcim_original_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_original_GBM19_Fw	0.001921	0.002122	4.312	8.193
rcim_original_HGBM19_Fw	0.002011	0.002217	4.493	10.617
rcim_original_ET19_Fw	0.002232	0.002432	4.985	11.357
rcim_original_XGBM19_Fw	0.002594	0.002814	5.805	10.574
paper_original_best_Fw	0.002769	0.002951	6.250	13.827
rcim_original_SVM19_Fw	0.003052	0.003324	6.767	13.827
rcim_original_ELM19_Fw	0.005423	0.005731	12.130	28.721
rcim_original_MLP19_Fw	0.018754	0.022589	42.943	107.515

Retuned Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
rcim_retuned_RF19_Fw	0.001487	0.001699	3.292	5.998
rcim_retuned_ERT19_Fw	0.001807	0.002010	4.039	7.599
paper_retuned_best_Fw	0.001839	0.002041	4.109	9.866
rcim_retuned_HGBM19_Fw	0.001851	0.002056	4.126	9.401
rcim_retuned_LGBM19_Fw	0.001851	0.002055	4.135	9.866
rcim_retuned_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_retuned_ET19_Fw	0.002001	0.002196	4.426	9.533
rcim_retuned_XGBM19_Fw	0.002054	0.002264	4.588	10.488
rcim_retuned_SVM19_Fw	0.003167	0.003428	7.035	12.971
rcim_retuned_ELM19_Fw	0.007182	0.007463	16.181	40.024
rcim_retuned_MLP19_Fw	0.016647	0.020154	38.510	86.197

RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
RF19_Fw	0.002164	0.002371	4.841	11.061
ERT19_Fw	0.002295	0.002491	5.163	13.141
GBM19_Fw	0.002779	0.002981	6.238	12.995
track1_best_Fw	0.003014	0.003204	6.819	11.638
DT19_Fw	0.003122	0.003313	7.011	13.928
SVM19_Fw	0.003236	0.003487	7.185	11.841
HGBM19_Fw	0.003251	0.003464	7.315	13.802
ET19_Fw	0.003267	0.003467	7.339	14.670
XGBM19_Fw	0.004190	0.004396	9.407	22.660
LGBM19_Fw	0.006812	0.007009	15.415	30.398
ELM19_Fw	0.007281	0.007573	16.352	36.326
MLP19_Fw	0.038690	0.047157	89.567	201.437

Wave 1 Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_global	0.002998	0.003364	6.590	11.601
tree_Fw	0.003053	0.003395	6.731	11.995
periodic_mlp_Fw	0.003366	0.003794	7.467	12.904
periodic_mlp_global	0.003380	0.003791	7.493	14.349
harmonic_regression_Fw	0.003394	0.003807	7.534	11.391
residual_harmonic_mlp_Fw	0.003434	0.003876	7.618	13.617
feedforward_Fw	0.003466	0.003904	7.685	13.132
feedforward_global	0.003550	0.003979	7.916	16.195
residual_harmonic_mlp_global	0.003649	0.004051	8.123	14.713
harmonic_regression_global	0.018314	0.018547	41.749	79.404

Wave 2.1 Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_lstm_sequence_global	0.002726	0.002959	6.142	14.092
periodic_gru_sequence_global	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_Fw	0.003186	0.003438	7.077	11.974
periodic_lstm_sequence_Fw	0.003266	0.003550	7.258	11.961
gru_sequence_Fw	0.003330	0.003762	7.378	13.029
periodic_temporal_convolution_Fw	0.003335	0.003708	7.404	12.518
lstm_sequence_Fw	0.003366	0.003800	7.450	11.807
periodic_temporal_convolution_global	0.003407	0.003724	7.581	14.480
lstm_sequence_global	0.003445	0.003863	7.642	12.032
temporal_convolution_global	0.003508	0.003928	7.792	14.045
gru_sequence_global	0.003546	0.003975	7.869	14.344
temporal_convolution_Fw	0.003603	0.004031	8.028	14.674

Wave 2.3 Residual Harmonic Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_Fw	0.003194	0.003499	7.083	13.041
residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.003229	0.003533	7.164	12.682
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003245	0.003562	7.192	12.555
residual_harmonic_gru_sequence_sparse_rcim_global	0.003287	0.003611	7.282	12.461
residual_harmonic_lstm_sequence_dense240_global	0.006247	0.008597	14.061	20.066
residual_harmonic_gru_sequence_dense240_global	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_Fw	0.006983	0.009275	15.722	19.139
residual_harmonic_lstm_sequence_dense240_Fw	0.007042	0.009370	15.868	19.358
residual_harmonic_lstm_sequence_dense360_Fw	0.007731	0.010235	17.430	21.285
residual_harmonic_gru_sequence_dense360_global	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_Fw	0.007869	0.010574	17.740	20.918
residual_harmonic_lstm_sequence_dense360_global	0.008760	0.012992	19.759	24.068

Wave 3.1 Offset-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_global	0.011772	0.013373	25.070	64.043

Wave 3.2 Harmonic-Offset Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_clean_sequential_residual_offset_global	0.014946	0.015784	31.986	76.955
track2f_bis_harmonic_residual_offset_global	0.039482	0.039637	84.805	170.090

Wave 3.3 Curve-Aware Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_full_curve_composite_global	0.001798	0.002176	3.621	8.564
track2g_curve_aware_raw_centered_shape_global	0.008354	0.008849	18.524	30.195
track2g_curve_aware_raw_offset_global	0.009768	0.010154	20.971	56.927
track2g_curve_aware_pointwise_control_global	0.015865	0.016241	34.752	79.981

Wave 5.2 series Robust-Loss Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_log_cosh_robust_global	0.006651	0.007133	14.636	27.123
track2h_mae_robust_global	0.007132	0.007593	15.736	28.173
track2h_smooth_l1_robust_global	0.034438	0.034618	74.024	167.540

Wave 5.2 series Quantile Probabilistic Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_gaussian_nll_global	0.016491	0.016807	35.270	86.670
track2h_quantile_p10_p50_p90_global	0.034013	0.034165	72.887	153.467

Wave 5.2 series Mixture Density Heads Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k3_global	0.008047	0.008537	17.449	50.984
track2h_mdn_k2_global	0.014253	0.014687	30.002	96.021

Wave 4.4 Latent-State Hysteresis Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_gru_offset_residual_global	0.002172	0.002635	4.462	8.584
track2h_l_causal_tcn_offset_residual_global	0.021679	0.022716	46.237	109.161

Wave 5.1 Harmonic-Prior Residual Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.001769	0.002131	3.577	8.089
wave3_harmonic_prior_residual_pointwise_control_global	0.004833	0.005371	10.455	20.017

Wave 5.2B Offset And Harmonic Guided Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_global	0.001929	0.002295	3.959	9.602

Polished Model-Development Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_gru_sequence_global	0.001350	0.001634	2.845	5.761
polished_periodic_lstm_sequence_global	0.001378	0.001675	2.880	5.456
polished_wave4_3_mixture_density_k3_global	0.001484	0.001821	3.068	7.052
polished_wave4_3_mixture_density_k2_global	0.001549	0.001885	3.211	7.142
polished_periodic_mlp_harmonic_global	0.001662	0.001991	3.332	8.068
polished_wave4_1_mae_robust_loss_global	0.001690	0.002042	3.386	8.042
polished_wave4_1_log_cosh_robust_loss_global	0.001697	0.002047	3.399	8.006
polished_wave4_2_quantile_p10_p50_p90_global	0.001706	0.002060	3.426	8.218
polished_wave3_3_raw_centered_shape_curve_aware_global	0.001735	0.002095	3.482	8.103
polished_wave4_1_smooth_l1_robust_loss_global	0.001747	0.002103	3.516	7.971
polished_wave3_2_harmonic_residual_offset_global	0.001750	0.002122	3.522	8.358
polished_wave3_3_curve_aware_pointwise_control_global	0.001750	0.002111	3.530	7.976
polished_wave4_2_gaussian_nll_global	0.001765	0.002131	3.557	8.116
polished_wave3_3_full_curve_composite_global	0.001780	0.002154	3.590	7.845
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.001790	0.002154	3.624	8.003
polished_wave3_3_raw_offset_curve_aware_global	0.001814	0.002186	3.666	8.584
polished_wave5_1_harmonic_prior_pointwise_control_global	0.001815	0.002180	3.676	8.087
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.001850	0.002239	3.739	8.348
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.001877	0.002267	3.809	8.541
polished_wave3_1_sequential_residual_offset_probe_global	0.002054	0.002500	4.197	8.280
polished_gru_sequence_global	0.002057	0.002506	4.205	8.244
polished_periodic_temporal_convolution_global	0.002089	0.002456	4.310	9.667
polished_wave3_2_clean_sequential_residual_offset_global	0.002106	0.002557	4.323	8.273
polished_lstm_sequence_global	0.002109	0.002568	4.324	8.291
polished_feedforward_global	0.002120	0.002583	4.350	8.438
polished_tree_global	0.002125	0.002612	4.355	8.534
polished_residual_harmonic_mlp_global	0.002129	0.002584	4.366	8.366
polished_wave4_4_gru_latent_offset_residual_global	0.002146	0.002616	4.410	8.445
polished_periodic_mlp_global	0.002152	0.002610	4.417	8.602
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.002162	0.002630	4.434	8.644
polished_temporal_convolution_global	0.002206	0.002674	4.557	8.314
polished_residual_harmonic_gru_sequence_dense240_global	0.002901	0.003732	6.161	8.815
polished_residual_harmonic_lstm_sequence_dense240_global	0.003106	0.004024	6.613	9.157
polished_harmonic_regression_global	0.003801	0.004289	8.202	14.746
polished_residual_harmonic_lstm_sequence_dense360_global	0.004376	0.006467	9.492	11.407

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_residual_harmonic_gru_sequence_dense360_global	0.004459	0.006784	9.683	11.885

Backward Comparison

Retuned Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Bw	0.002766	0.003300	5.398	12.280
rcim_retuned_ET19_Bw	0.003441	0.004028	7.021	15.287
rcim_retuned_ERT19_Bw	0.003551	0.004161	7.269	13.187
rcim_retuned_RF19_Bw	0.003649	0.004256	7.543	15.083
paper_retuned_best_Bw	0.003675	0.004284	7.572	15.645
rcim_retuned_SVM19_Bw	0.004016	0.004599	8.813	17.215
rcim_retuned_DT19_Bw	0.004578	0.005169	9.728	19.601
rcim_retuned_HGBM19_Bw	0.004683	0.005301	9.978	17.712
rcim_retuned_LGBM19_Bw	0.008105	0.008655	18.057	35.748
rcim_retuned_ELM19_Bw	0.008917	0.009518	20.169	51.896
rcim_retuned_XGBM19_Bw	0.010679	0.011209	24.184	48.082
rcim_retuned_MLP19_Bw	0.019115	0.023025	44.141	88.991

RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
SVM19_Bw	0.004822	0.005116	10.864	25.533
track1_best_Bw	0.005027	0.005212	11.860	48.106
LGBM19_Bw	0.005037	0.005231	11.880	48.106
GBM19_Bw	0.005180	0.005363	12.252	49.984
DT19_Bw	0.005226	0.005409	12.359	48.860
ERT19_Bw	0.005258	0.005442	12.434	51.665
RF19_Bw	0.005392	0.005584	12.731	55.740
ET19_Bw	0.006273	0.006520	14.314	48.624
HGBM19_Bw	0.006619	0.006834	15.494	53.982
XGBM19_Bw	0.007991	0.008195	18.722	59.067
ELM19_Bw	0.010071	0.010486	23.034	52.159
MLP19_Bw	0.036681	0.044921	86.544	225.831

Wave 1 Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_Bw	0.003258	0.003651	7.051	14.116
tree_global	0.003290	0.003702	7.118	13.703
periodic_mlp_Bw	0.003574	0.004006	7.807	13.970
periodic_mlp_global	0.003610	0.004030	7.879	13.661
residual_harmonic_mlp_global	0.003639	0.004044	7.967	14.489
feedforward_Bw	0.003644	0.004067	7.977	14.284
residual_harmonic_mlp_Bw	0.003672	0.004089	8.046	14.973
harmonic_regression_Bw	0.003745	0.004173	8.171	15.684
feedforward_global	0.003741	0.004188	8.186	15.259
harmonic_regression_global	0.018006	0.018276	41.275	83.603

Wave 2.1 Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
periodic_gru_sequence_global	0.002630	0.002876	6.010	12.693
periodic_lstm_sequence_Bw	0.002625	0.002877	6.013	15.382
periodic_lstm_sequence_global	0.002689	0.002956	6.098	14.674
lstm_sequence_global	0.003515	0.003944	7.666	12.502
lstm_sequence_Bw	0.003555	0.003985	7.767	14.507
gru_sequence_Bw	0.003626	0.004082	7.907	12.900
periodic_temporal_convolution_global	0.003604	0.003948	7.935	13.804
gru_sequence_global	0.003637	0.004080	7.946	14.580
periodic_temporal_convolution_Bw	0.003628	0.003987	7.979	13.839
temporal_convolution_Bw	0.003742	0.004166	8.184	13.908
temporal_convolution_global	0.003994	0.004438	8.798	18.339

Wave 2.3 Residual Harmonic Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.003440	0.003793	7.510	13.160
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003491	0.003876	7.627	13.540
residual_harmonic_gru_sequence_sparse_rcim_Bw	0.003502	0.003857	7.654	12.829
residual_harmonic_gru_sequence_sparse_rcim_global	0.003584	0.003957	7.859	14.946
residual_harmonic_lstm_sequence_dense240_global	0.006591	0.008932	14.858	21.728
residual_harmonic_gru_sequence_dense240_global	0.006786	0.009221	15.282	20.990
residual_harmonic_lstm_sequence_dense240_Bw	0.007367	0.009945	16.660	22.200
residual_harmonic_gru_sequence_dense360_global	0.008181	0.011621	18.502	23.968
residual_harmonic_lstm_sequence_dense360_global	0.008860	0.013061	20.073	23.940
residual_harmonic_gru_sequence_dense240_Bw	0.008984	0.012987	20.358	26.370
residual_harmonic_gru_sequence_dense360_Bw	0.009370	0.013165	21.267	25.901
residual_harmonic_lstm_sequence_dense360_Bw	0.010268	0.014769	23.355	29.779

Wave 3.1 Offset-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_global	0.008806	0.009398	18.279	32.785

Wave 3.2 Harmonic-Offset Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_clean_sequential_residual_offset_global	0.006660	0.007243	13.756	24.509
track2f_bis_harmonic_residual_offset_global	0.008474	0.008976	17.214	35.408

Wave 3.3 Curve-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_full_curve_composite_global	0.002258	0.002737	4.040	10.132
track2g_curve_aware_raw_offset_global	0.009653	0.010130	20.050	31.531
track2g_curve_aware_raw_centered_shape_global	0.011137	0.011575	23.462	35.847
track2g_curve_aware_pointwise_control_global	0.013358	0.013773	27.446	48.301

Wave 5.2 series Robust-Loss Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_log_cosh_robust_global	0.011005	0.011437	23.172	35.371
track2h_mae_robust_global	0.012719	0.013122	26.537	43.911
track2h_smooth_l1_robust_global	0.014760	0.015146	30.600	49.591

Wave 5.2 series Quantile Probabilistic Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_gaussian_nll_global	0.009465	0.009894	19.945	35.904
track2h_quantile_p10_p50_p90_global	0.011361	0.011796	23.554	42.592

Wave 5.2 series Mixture Density Heads Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k2_global	0.011621	0.012094	24.024	46.001
track2h_mdn_k3_global	0.016585	0.016922	34.616	54.718

Wave 4.4 Latent-State Hysteresis Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_gru_offset_residual_global	0.002532	0.003071	4.689	10.217
track2h_l_causal_tcn_offset_residual_global	0.021480	0.023284	46.491	83.800

Wave 5.1 Harmonic-Prior Residual Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.002581	0.003040	4.545	9.839
wave3_harmonic_prior_residual_pointwise_control_global	0.009539	0.010032	19.825	33.810

Wave 5.2B Offset And Harmonic Guided Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_global	0.002533	0.002985	4.422	10.180

Polished Model-Development Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_lstm_sequence_global	0.001222	0.001520	2.389	4.595
polished_periodic_gru_sequence_global	0.001204	0.001497	2.413	5.333
polished_wave4_3_mixture_density_k3_global	0.001693	0.002077	3.166	7.127
polished_wave4_3_mixture_density_k2_global	0.001980	0.002392	3.505	9.704
polished_wave4_2_quantile_p10_p50_p90_global	0.002124	0.002578	3.763	9.707
polished_wave3_2_harmonic_residual_offset_global	0.002134	0.002590	3.799	9.605
polished_wave4_1_mae_robust_loss_global	0.002137	0.002598	3.803	9.826
polished_wave4_1_log_cosh_robust_loss_global	0.002163	0.002604	3.833	9.703
polished_wave3_3_curve_aware_pointwise_control_global	0.002197	0.002649	3.920	9.686
polished_wave3_3_raw_centered_shape_curve_aware_global	0.002235	0.002667	3.952	10.367
polished_wave3_3_raw_offset_curve_aware_global	0.002217	0.002687	3.992	9.699
polished_wave4_2_gaussian_nll_global	0.002261	0.002722	4.008	10.676
polished_wave4_1_smooth_l1_robust_loss_global	0.002306	0.002752	4.144	9.959
polished_wave3_3_full_curve_composite_global	0.002292	0.002794	4.165	9.742
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.002302	0.002808	4.172	9.983
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.002472	0.002914	4.267	10.107
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.002351	0.002847	4.271	9.883
polished_wave5_1_harmonic_prior_pointwise_control_global	0.002513	0.002956	4.347	10.558
polished_periodic_mlp_harmonic_global	0.002471	0.002952	4.363	10.739
polished_gru_sequence_global	0.002424	0.002952	4.428	10.133
polished_lstm_sequence_global	0.002434	0.002949	4.461	10.188
polished_wave3_2_clean_sequential_residual_offset_global	0.002457	0.002976	4.476	10.158
polished_wave3_1_sequential_residual_offset_probe_global	0.002464	0.002989	4.508	10.147
polished_wave4_4_gru_latent_offset_residual_global	0.002477	0.003011	4.535	10.117
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.002489	0.003045	4.557	10.219
polished_periodic_temporal_convolution_global	0.002561	0.003049	4.759	10.768
polished_periodic_mlp_global	0.002708	0.003255	4.808	10.597
polished_feedforward_global	0.002706	0.003251	4.815	10.600
polished_residual_harmonic_mlp_global	0.002711	0.003257	4.832	10.598
polished_temporal_convolution_global	0.002601	0.003160	4.856	10.225
polished_tree_global	0.002756	0.003287	4.934	10.752
polished_residual_harmonic_gru_sequence_dense240_global	0.003355	0.004270	6.644	10.393
polished_residual_harmonic_lstm_sequence_dense240_global	0.003566	0.004593	7.128	10.806
polished_harmonic_regression_global	0.004168	0.004730	8.121	14.643
polished_residual_harmonic_gru_sequence_dense360_global	0.004814	0.007204	9.980	12.623

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_residual_harmonic_lstm_sequence_dense360_global	0.004876	0.006971	10.030	12.714

Global Model Direction Breakdown

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
feedforward_global	forward	0.003550	0.003979	7.916	16.195
feedforward_global	backward	0.003741	0.004188	8.186	15.259
feedforward_global	combined	0.003646	0.004083	8.051	15.583
gru_sequence_global	forward	0.003546	0.003975	7.869	14.344
gru_sequence_global	backward	0.003637	0.004080	7.946	14.580
gru_sequence_global	combined	0.003591	0.004028	7.907	14.389
harmonic_regression_global	forward	0.018314	0.018547	41.749	79.404
harmonic_regression_global	backward	0.018006	0.018276	41.275	83.603
harmonic_regression_global	combined	0.018160	0.018412	41.512	82.048
lstm_sequence_global	forward	0.003445	0.003863	7.642	12.032
lstm_sequence_global	backward	0.003515	0.003944	7.666	12.502
lstm_sequence_global	combined	0.003480	0.003903	7.654	12.430
periodic_gru_sequence_global	forward	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_global	backward	0.002630	0.002876	6.010	12.693
periodic_gru_sequence_global	combined	0.002704	0.002949	6.139	13.200
periodic_lstm_sequence_global	forward	0.002726	0.002959	6.142	14.092
periodic_lstm_sequence_global	backward	0.002689	0.002956	6.098	14.674
periodic_lstm_sequence_global	combined	0.002707	0.002958	6.120	14.717
periodic_mlp_global	forward	0.003380	0.003791	7.493	14.349
periodic_mlp_global	backward	0.003610	0.004030	7.879	13.661
periodic_mlp_global	combined	0.003495	0.003910	7.686	14.284
periodic_temporal_convolution_global	forward	0.003407	0.003724	7.581	14.480
periodic_temporal_convolution_global	backward	0.003604	0.003948	7.935	13.804
periodic_temporal_convolution_global	combined	0.003506	0.003836	7.758	14.308
polished_feedforward_global	forward	0.002120	0.002583	4.350	8.438
polished_feedforward_global	backward	0.002706	0.003251	4.815	10.600
polished_feedforward_global	combined	0.002404	0.002907	4.575	10.056
polished_gru_sequence_global	forward	0.002057	0.002506	4.205	8.244
polished_gru_sequence_global	backward	0.002424	0.002952	4.428	10.133
polished_gru_sequence_global	combined	0.002235	0.002722	4.313	9.564
polished_harmonic_regression_global	forward	0.003801	0.004289	8.202	14.746
polished_harmonic_regression_global	backward	0.004168	0.004730	8.121	14.643
polished_harmonic_regression_global	combined	0.003979	0.004503	8.163	14.763
polished_lstm_sequence_global	forward	0.002109	0.002568	4.324	8.291
polished_lstm_sequence_global	backward	0.002434	0.002949	4.461	10.188
polished_lstm_sequence_global	combined	0.002266	0.002753	4.390	9.554
polished_periodic_gru_sequence_global	forward	0.001350	0.001634	2.845	5.761
polished_periodic_gru_sequence_global	backward	0.001204	0.001497	2.413	5.333
polished_periodic_gru_sequence_global	combined	0.001279	0.001568	2.636	5.690

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_lstm_sequence_global	forward	0.001378	0.001675	2.880	5.456
polished_periodic_lstm_sequence_global	backward	0.001222	0.001520	2.389	4.595
polished_periodic_lstm_sequence_global	combined	0.001303	0.001600	2.642	5.314
polished_periodic_mlp_global	forward	0.002152	0.002610	4.417	8.602
polished_periodic_mlp_global	backward	0.002708	0.003255	4.808	10.597
polished_periodic_mlp_global	combined	0.002421	0.002922	4.606	9.853
polished_periodic_mlp_harmonic_global	forward	0.001662	0.001991	3.332	8.068
polished_periodic_mlp_harmonic_global	backward	0.002471	0.002952	4.363	10.739
polished_periodic_mlp_harmonic_global	combined	0.002054	0.002457	3.832	9.502
polished_periodic_temporal_convolution_global	forward	0.002089	0.002456	4.310	9.667
polished_periodic_temporal_convolution_global	backward	0.002561	0.003049	4.759	10.768
polished_periodic_temporal_convolution_global	combined	0.002318	0.002743	4.527	9.986
polished_residual_harmonic_gru_sequence_dense240_global	forward	0.002901	0.003732	6.161	8.815
polished_residual_harmonic_gru_sequence_dense240_global	backward	0.003355	0.004270	6.644	10.393
polished_residual_harmonic_gru_sequence_dense240_global	combined	0.003121	0.003993	6.395	10.129
polished_residual_harmonic_gru_sequence_dense360_global	forward	0.004459	0.006784	9.683	11.885
polished_residual_harmonic_gru_sequence_dense360_global	backward	0.004814	0.007204	9.980	12.623
polished_residual_harmonic_gru_sequence_dense360_global	combined	0.004631	0.006987	9.827	12.042
polished_residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.001877	0.002267	3.809	8.541
polished_residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.002351	0.002847	4.271	9.883
polished_residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.002107	0.002548	4.033	9.587
polished_residual_harmonic_lstm_sequence_dense240_global	forward	0.003106	0.004024	6.613	9.157
polished_residual_harmonic_lstm_sequence_dense240_global	backward	0.003566	0.004593	7.128	10.806
polished_residual_harmonic_lstm_sequence_dense240_global	combined	0.003329	0.004300	6.862	10.535
polished_residual_harmonic_lstm_sequence_dense360_global	forward	0.004376	0.006467	9.492	11.407
polished_residual_harmonic_lstm_sequence_dense360_global	backward	0.004876	0.006971	10.030	12.714
polished_residual_harmonic_lstm_sequence_dense360_global	combined	0.004618	0.006711	9.753	12.396
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.001850	0.002239	3.739	8.348
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.002302	0.002808	4.172	9.983
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.002069	0.002515	3.949	9.574
polished_residual_harmonic_mlp_global	forward	0.002129	0.002584	4.366	8.366
polished_residual_harmonic_mlp_global	backward	0.002711	0.003257	4.832	10.598
polished_residual_harmonic_mlp_global	combined	0.002411	0.002910	4.592	10.130
polished_temporal_convolution_global	forward	0.002206	0.002674	4.557	8.314

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_temporal_convolution_global	backward	0.002601	0.003160	4.856	10.225
polished_temporal_convolution_global	combined	0.002398	0.002909	4.702	9.713
polished_tree_global	forward	0.002125	0.002612	4.355	8.534
polished_tree_global	backward	0.002756	0.003287	4.934	10.752
polished_tree_global	combined	0.002431	0.002939	4.635	9.931
polished_wave3_1_sequential_residual_offset_probe_global	forward	0.002054	0.002500	4.197	8.280
polished_wave3_1_sequential_residual_offset_probe_global	backward	0.002464	0.002989	4.508	10.147
polished_wave3_1_sequential_residual_offset_probe_global	combined	0.002252	0.002737	4.348	9.622
polished_wave3_2_clean_sequential_residual_offset_global	forward	0.002106	0.002557	4.323	8.273
polished_wave3_2_clean_sequential_residual_offset_global	backward	0.002457	0.002976	4.476	10.158
polished_wave3_2_clean_sequential_residual_offset_global	combined	0.002276	0.002760	4.397	9.579
polished_wave3_2_harmonic_residual_offset_global	forward	0.001750	0.002122	3.522	8.358
polished_wave3_2_harmonic_residual_offset_global	backward	0.002134	0.002590	3.799	9.605
polished_wave3_2_harmonic_residual_offset_global	combined	0.001936	0.002349	3.656	9.145
polished_wave3_3_curve_aware_pointwise_control_global	forward	0.001750	0.002111	3.530	7.976
polished_wave3_3_curve_aware_pointwise_control_global	backward	0.002197	0.002649	3.920	9.686
polished_wave3_3_curve_aware_pointwise_control_global	combined	0.001967	0.002372	3.719	9.186
polished_wave3_3_full_curve_composite_global	forward	0.001780	0.002154	3.590	7.845
polished_wave3_3_full_curve_composite_global	backward	0.002292	0.002794	4.165	9.742
polished_wave3_3_full_curve_composite_global	combined	0.002028	0.002464	3.869	9.205
polished_wave3_3_raw_centered_shape_curve_aware_global	forward	0.001735	0.002095	3.482	8.103
polished_wave3_3_raw_centered_shape_curve_aware_global	backward	0.002235	0.002667	3.952	10.367
polished_wave3_3_raw_centered_shape_curve_aware_global	combined	0.001977	0.002372	3.710	9.358
polished_wave3_3_raw_offset_curve_aware_global	forward	0.001814	0.002186	3.666	8.584
polished_wave3_3_raw_offset_curve_aware_global	backward	0.002217	0.002687	3.992	9.699
polished_wave3_3_raw_offset_curve_aware_global	combined	0.002009	0.002428	3.824	9.323
polished_wave4_1_log_cosh_robust_loss_global	forward	0.001697	0.002047	3.399	8.006
polished_wave4_1_log_cosh_robust_loss_global	backward	0.002163	0.002604	3.833	9.703
polished_wave4_1_log_cosh_robust_loss_global	combined	0.001923	0.002317	3.610	9.289
polished_wave4_1_mae_robust_loss_global	forward	0.001690	0.002042	3.386	8.042
polished_wave4_1_mae_robust_loss_global	backward	0.002137	0.002598	3.803	9.826
polished_wave4_1_mae_robust_loss_global	combined	0.001907	0.002311	3.588	9.258
polished_wave4_1_smooth_l1_robust_loss_global	forward	0.001747	0.002103	3.516	7.971
polished_wave4_1_smooth_l1_robust_loss_global	backward	0.002306	0.002752	4.144	9.959

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_wave4_1_smooth_l1_robust_loss_global	combined	0.002018	0.002417	3.821	9.619
polished_wave4_2_gaussian_nll_global	forward	0.001765	0.002131	3.557	8.116
polished_wave4_2_gaussian_nll_global	backward	0.002261	0.002722	4.008	10.676
polished_wave4_2_gaussian_nll_global	combined	0.002006	0.002417	3.776	9.403
polished_wave4_2_quantile_p10_p50_p90_global	forward	0.001706	0.002060	3.426	8.218
polished_wave4_2_quantile_p10_p50_p90_global	backward	0.002124	0.002578	3.763	9.707
polished_wave4_2_quantile_p10_p50_p90_global	combined	0.001908	0.002311	3.589	9.208
polished_wave4_3_mixture_density_k2_global	forward	0.001549	0.001885	3.211	7.142
polished_wave4_3_mixture_density_k2_global	backward	0.001980	0.002392	3.505	9.704
polished_wave4_3_mixture_density_k2_global	combined	0.001758	0.002131	3.354	8.461
polished_wave4_3_mixture_density_k3_global	forward	0.001484	0.001821	3.068	7.052
polished_wave4_3_mixture_density_k3_global	backward	0.001693	0.002077	3.166	7.127
polished_wave4_3_mixture_density_k3_global	combined	0.001585	0.001945	3.116	7.127
polished_wave4_4_causal_tcn_latent_offset_residual_global	forward	0.002162	0.002630	4.434	8.644
polished_wave4_4_causal_tcn_latent_offset_residual_global	backward	0.002489	0.003045	4.557	10.219
polished_wave4_4_causal_tcn_latent_offset_residual_global	combined	0.002321	0.002831	4.493	9.907
polished_wave4_4_gru_latent_offset_residual_global	forward	0.002146	0.002616	4.410	8.445
polished_wave4_4_gru_latent_offset_residual_global	backward	0.002477	0.003011	4.535	10.117
polished_wave4_4_gru_latent_offset_residual_global	combined	0.002306	0.002807	4.470	9.612
polished_wave5_1_harmonic_prior_pointwise_control_global	forward	0.001815	0.002180	3.676	8.087
polished_wave5_1_harmonic_prior_pointwise_control_global	backward	0.002513	0.002956	4.347	10.558
polished_wave5_1_harmonic_prior_pointwise_control_global	combined	0.002153	0.002556	4.001	9.336
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	forward	0.001790	0.002154	3.624	8.003
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	backward	0.002472	0.002914	4.267	10.107
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	combined	0.002121	0.002522	3.936	9.529
residual_harmonic_gru_sequence_dense240_global	forward	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_global	backward	0.006786	0.009221	15.282	20.990
residual_harmonic_gru_sequence_dense240_global	combined	0.006660	0.009090	15.007	20.940
residual_harmonic_gru_sequence_dense360_global	forward	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_global	backward	0.008181	0.011621	18.502	23.968
residual_harmonic_gru_sequence_dense360_global	combined	0.008012	0.011416	18.090	23.559
residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.003287	0.003611	7.282	12.461
residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.003584	0.003957	7.859	14.946

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.003435	0.003784	7.571	13.869
residual_harmonic_lstm_sequence_dense240_global	forward	0.006247	0.008597	14.061	20.066
residual_harmonic_lstm_sequence_dense240_global	backward	0.006591	0.008932	14.858	21.728
residual_harmonic_lstm_sequence_dense240_global	combined	0.006419	0.008765	14.460	20.628
residual_harmonic_lstm_sequence_dense360_global	forward	0.008760	0.012992	19.759	24.068
residual_harmonic_lstm_sequence_dense360_global	backward	0.008860	0.013061	20.073	23.940
residual_harmonic_lstm_sequence_dense360_global	combined	0.008810	0.013026	19.916	24.052
residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.003245	0.003562	7.192	12.555
residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.003491	0.003876	7.627	13.540
residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.003368	0.003719	7.409	13.041
residual_harmonic_mlp_global	forward	0.003649	0.004051	8.123	14.713
residual_harmonic_mlp_global	backward	0.003639	0.004044	7.967	14.489
residual_harmonic_mlp_global	combined	0.003644	0.004048	8.045	14.717
sequential_residual_offset_probe_global	forward	0.011772	0.013373	25.070	64.043
sequential_residual_offset_probe_global	backward	0.008806	0.009398	18.279	32.785
sequential_residual_offset_probe_global	combined	0.010335	0.011447	21.780	54.389
temporal_convolution_global	forward	0.003508	0.003928	7.792	14.045
temporal_convolution_global	backward	0.003994	0.004438	8.798	18.339
temporal_convolution_global	combined	0.003751	0.004183	8.295	17.247
track2f_bis_clean_sequential_residual_offset_global	forward	0.014946	0.015784	31.986	76.955
track2f_bis_clean_sequential_residual_offset_global	backward	0.006660	0.007243	13.756	24.509
track2f_bis_clean_sequential_residual_offset_global	combined	0.010931	0.011645	23.153	66.991
track2f_bis_harmonic_residual_offset_global	forward	0.039482	0.039637	84.805	170.090
track2f_bis_harmonic_residual_offset_global	backward	0.008474	0.008976	17.214	35.408
track2f_bis_harmonic_residual_offset_global	combined	0.024458	0.024780	52.055	157.788
track2g_curve_aware_full_curve_composite_global	forward	0.001798	0.002176	3.621	8.564
track2g_curve_aware_full_curve_composite_global	backward	0.002258	0.002737	4.040	10.132
track2g_curve_aware_full_curve_composite_global	combined	0.002021	0.002448	3.824	9.460
track2g_curve_aware_pointwise_control_global	forward	0.015865	0.016241	34.752	79.981
track2g_curve_aware_pointwise_control_global	backward	0.013358	0.013773	27.446	48.301
track2g_curve_aware_pointwise_control_global	combined	0.014650	0.015045	31.212	69.019
track2g_curve_aware_raw_centered_shape_global	forward	0.008354	0.008849	18.524	30.195
track2g_curve_aware_raw_centered_shape_global	backward	0.011137	0.011575	23.462	35.847
track2g_curve_aware_raw_centered_shape_global	combined	0.009702	0.010170	20.917	33.589
track2g_curve_aware_raw_offset_global	forward	0.009768	0.010154	20.971	56.927

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_offset_global	backward	0.009653	0.010130	20.050	31.531
track2g_curve_aware_raw_offset_global	combined	0.009712	0.010143	20.524	49.082
track2h_gaussian_nll_global	forward	0.016491	0.016807	35.270	86.670
track2h_gaussian_nll_global	backward	0.009465	0.009894	19.945	35.904
track2h_gaussian_nll_global	combined	0.013087	0.013458	27.845	77.071
track2h_l_causal_tcn_offset_residual_global	forward	0.021679	0.022716	46.237	109.161
track2h_l_causal_tcn_offset_residual_global	backward	0.021480	0.023284	46.491	83.800
track2h_l_causal_tcn_offset_residual_global	combined	0.021583	0.022991	46.360	97.912
track2h_l_gru_offset_residual_global	forward	0.002172	0.002635	4.462	8.584
track2h_l_gru_offset_residual_global	backward	0.002532	0.003071	4.689	10.217
track2h_l_gru_offset_residual_global	combined	0.002346	0.002846	4.572	9.808
track2h_log_cosh_robust_global	forward	0.006651	0.007133	14.636	27.123
track2h_log_cosh_robust_global	backward	0.011005	0.011437	23.172	35.371
track2h_log_cosh_robust_global	combined	0.008761	0.009218	18.772	32.617
track2h_mae_robust_global	forward	0.007132	0.007593	15.736	28.173
track2h_mae_robust_global	backward	0.012719	0.013122	26.537	43.911
track2h_mae_robust_global	combined	0.009839	0.010272	20.969	37.437
track2h_mdn_k2_global	forward	0.014253	0.014687	30.002	96.021
track2h_mdn_k2_global	backward	0.011621	0.012094	24.024	46.001
track2h_mdn_k2_global	combined	0.012978	0.013430	27.105	80.630
track2h_mdn_k3_global	forward	0.008047	0.008537	17.449	50.984
track2h_mdn_k3_global	backward	0.016585	0.016922	34.616	54.718
track2h_mdn_k3_global	combined	0.012184	0.012600	25.767	54.085
track2h_quantile_p10_p50_p90_global	forward	0.034013	0.034165	72.887	153.467
track2h_quantile_p10_p50_p90_global	backward	0.011361	0.011796	23.554	42.592
track2h_quantile_p10_p50_p90_global	combined	0.023037	0.023327	48.983	142.345
track2h_smooth_l1_robust_global	forward	0.034438	0.034618	74.024	167.540
track2h_smooth_l1_robust_global	backward	0.014760	0.015146	30.600	49.591
track2h_smooth_l1_robust_global	combined	0.024904	0.025183	52.984	152.879
tree_global	forward	0.002998	0.003364	6.590	11.601
tree_global	backward	0.003290	0.003702	7.118	13.703
tree_global	combined	0.003144	0.003533	6.854	13.314
wave3_harmonic_prior_residual_pointwise_control_global	forward	0.004833	0.005371	10.455	20.017
wave3_harmonic_prior_residual_pointwise_control_global	backward	0.009539	0.010032	19.825	33.810
wave3_harmonic_prior_residual_pointwise_control_global	combined	0.007113	0.007629	14.995	31.569
wave3_harmonic_prior_residual_smooth_l1_structured_global	forward	0.001769	0.002131	3.577	8.089
wave3_harmonic_prior_residual_smooth_l1_structured_global	backward	0.002581	0.003040	4.545	9.839
wave3_harmonic_prior_residual_smooth_l1_structured_global	combined	0.002163	0.002571	4.046	9.635
wave52b_offset_centered_shape_harmonic_global	forward	0.001929	0.002295	3.959	9.602
wave52b_offset_centered_shape_harmonic_global	backward	0.002533	0.002985	4.422	10.180
wave52b_offset_centered_shape_harmonic_global	combined	0.002221	0.002629	4.184	9.818

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-04-17-24-25__track2_full_directional_family_matrix_track2_dataset_surface_polished_dataset_global_2026_07_04\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-04-17-24-25__track2_full_directional_family_matrix_track2_dataset_surface_polished_dataset_global_2026_07_04\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.